gpeyt1-36 · Rankine thrust with water table

GEO PE · Long Calculation Problems I — Topic 8: Retaining Structures

Problem

A 22 ft high vertical-backed wall retains level granular backfill (c = 0, φ = 32°). Because the weep holes are clogged, a water table has built up 8 ft below the top of the backfill. γ = 118 pcf above the water table, γsat = 126 pcf below. Ignore wall friction; the wall moves enough to mobilize active pressure. The concrete stem is 18 in. thick.

Most nearly, the total horizontal force (soil + water) per foot of wall is:

A. 7,136 lb/ft
B. 13,251 lb/ft
C. 9,015 lb/ft
D. 8,774 lb/ft

Given equations / data

> $K_a = \tan^2(45 - \varphi/2)$; $\sigma'_h = K_a \sigma'_v$; $u = \gamma_w h_w$; P = area of pressure diagram

Worked solution

1. $K_a = 0.3073$
2. σ'_h at 8 ft = $0.3073 \times 118 \times 8 = 290$ psf
3. σ'_h at 22 ft = $290 + 0.3073 \times 63.6 \times 14 = 564$ psf
4. Soil: $\tfrac{1}{2}(290)(8) + 290(14) + \tfrac{1}{2}(274)(14) = 7{,}136$ lb/ft
5. Water: $\tfrac{1}{2}(62.4)(14)^2 = 6{,}115$ lb/ft
6. Total = 13,251 lb/ft

Final answer

13,251 lb/ft

Why the other values are wrong

- **7,136 lb/ft** — forgot the water pressure
- **9,015 lb/ft** — used γsat (total) below the WT and no separate water
- **8,774 lb/ft** — treated the backfill as dry

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.1.2 Rankine Earth Coefficients, p. 72

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gpeyt1-37 · Overturning FS

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Problem

A 12 ft tall plain-concrete gravity wall (γc = 150 pcf) has a vertical back face and a battered front face. Crest width 2 ft, base width 7 ft (toe at the front). Backfill is level, dry sand: γ = 120 pcf, φ = 30°, c = 0, no wall friction. Passive resistance in front is neglected. The foundation is dense gravel with δ = 28°.

Most nearly, the factor of safety against overturning is:

A. 2.12
B. 0.96
C. 3.18
D. 1.59

Given equations / data

> Ka = tan²(45 − φ/2); Pa = ½KaγH² at H/3; FSOT = ΣMresisting/ΣMoverturning about the toe

Worked solution

1. Ka = 1/3; Pa = ½(1/3)(120)(12)² = 2,880 lb/ft at 4.0 ft
2. Rectangle (back): W1 = 2×12×150 = 3,600 lb at 6.00 ft from toe
3. Triangle (front): W2 = ½×5×12×150 = 4,500 lb at 3.33 ft
4. MR = 36,600; MO = 2,880×4.0 = 11,520 ft-lb/ft
5. FSOT = 3.18

Final answer

3.18

Why the other values are wrong

- **2.12** — applied Pa at H/2
- **0.96** — took moments about the heel-side distances (wrong lever arms)
- **1.59** — forgot the ½ in Pa

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.1.2 Rankine; §3.18 / wall stability figures (moments about toe), p. 209

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gpeyt1-38 · Sliding FS with passive

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Problem

A cantilever wall has total vertical load (concrete + soil on heel) ΣV = 15,400 lb/ft and horizontal active thrust Ph = 6,200 lb/ft. The base is 9 ft wide and cast on sand with base friction δb = 26° (ca = 0). Soil in front of the toe is 3 ft deep (γ = 120 pcf, φ = 30°) and the designer will count full Rankine passive resistance on the toe. Overturning FS was already checked at 2.4.

Most nearly, the factor of safety against sliding is:

A. 1.21
B. 1.47
C. 1.24
D. 1.73

Given equations / data

> FSs = [(ΣV) tan δb + ca B + Pp]/Ph; Kp = tan²(45 + φ/2); Pp = ½Kpγ Df²

Worked solution

1. Base friction = 15,400 × tan26 = 7,511 lb/ft
2. Kp = 3.00; Pp = ½(3.00)(120)(3)² = 1,620 lb/ft
3. FSs = (7,511 + 1,620)/6,200 = 1.47

Final answer

1.47

Why the other values are wrong

- **1.21** — ignored passive resistance
- **1.24** — used Ka instead of Kp in front
- **1.73** — forgot the ½ in Pp

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 Factor of Safety against Sliding (cantilever/counterfort wall figure), p. 209; §3.1.2 Rankine

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gpeyt1-39 · Uniform surcharge thrust

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Problem

A 16 ft basement-type wall that can yield to the active state retains dry, level silty sand (γ = 115 pcf, φ = 30°, c = 0). A uniform surcharge of q = 400 psf from a parking area acts over the full backfill. No water; ignore wall friction. The designer needs the total thrust and its height above the base for the moment check.

Most nearly, the total thrust and its height above the base are:

A. 7,040 lb/ft at 6.1 ft
B. 7,040 lb/ft at 5.3 ft
C. 11,307 lb/ft at 6.8 ft
D. 5,973 lb/ft at 5.3 ft

Given equations / data

> $\sigma h = K_a(q + \gamma z)$; $P_q = K_a q H$ at $H/2$; $P_s = \frac{1}{2} K_a \gamma H^2$ at $H/3$

Worked solution

1. $K_a = 1/3$
2. $P_q = (1/3)(400)(16) = 2{,}133$ lb/ft at 8 ft
3. $P_s = \frac{1}{2}(1/3)(115)(16)^2 = 4{,}907$ lb/ft at 5.33 ft
4. $P = 7{,}040$ lb/ft; $\bar{y} = (2{,}133 \times 8 + 4{,}907 \times 5.33)/7{,}040 = 6.14$ ft

Final answer

7,040 lb/ft at 6.1 ft

Why the other values are wrong

- **7,040 lb/ft at 5.3 ft** — placed the whole resultant at H/3
- **11,307 lb/ft at 6.8 ft** — applied full q (forgot Ka on the surcharge)
- **5,973 lb/ft at 5.3 ft** — treated the surcharge as triangular

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.1.2 Rankine; §3.13.5.1 $\sigma_{hq} = K_a q$; §3.1.4 Load Distribution from Surcharge

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gpeyt1-40 · Wall base pressure

GEO PE · Long Calculation Problems I — Topic 8: Retaining Structures

Problem

For a cantilever retaining wall on medium dense sand, the resultant of all vertical forces ΣV = 18,200 lb/ft (Pv included) intersects the 10 ft base at d = 4.1 ft from the toe, computed from moments about the toe. The allowable bearing pressure for the sand is 4,000 psf. Overturning FS = 2.3, sliding FS = 1.6.

Most nearly, the maximum toe pressure qmax is:

A. 2,220 psf
B. 2,803 psf
C. 837 psf
D. 1,984 psf

Given equations / data

> `e = B/2 − d; qmax = (ΣV/B)(1 + 6e/B); qeq (Meyerhof) = ΣV/B′, B′ = B − 2e; for soils e ≤ B/6`

Worked solution

1. `e = 10/2 − 4.1 = 0.90 ft ≤ B/6 = 1.67 ✓`
2. `qmax = (18,200/10)(1 + 6×0.90/10) = 2,803 psf`
3. `qmin = 837 psf; qeq = 18,200/8.2 = 2,220 psf (for comparison)`

Final answer

2,803 psf

Why the other values are wrong

- **2,220 psf** — reported Meyerhof uniform qeq
- **837 psf** — reported qmin (heel)
- **1,984 psf** — used (1 + e/B), dropped the 6

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 Applied Stress at Base (qmax, qmin, qeq) and eccentricity criteria (cantilever wall figure), p. 209

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